

Data Structure(CSL202)

Dr. Sk Subidh Ali (subidh@iitbhilai.ac.in)



Breadth First Search

Breadth First Search

- Visits nodes level by level starting from the source node.
- Processes all neighbors first, before moving deeper.
- Uses a queue to manage nodes to be explored.
- Ensures each node is processed only once.

Breadth First Search

Algorithm:

1. Start from a source node.
2. Visit all its neighbors.
3. Move to neighbors' neighbors.
4. Repeat until all reachable nodes are visited.

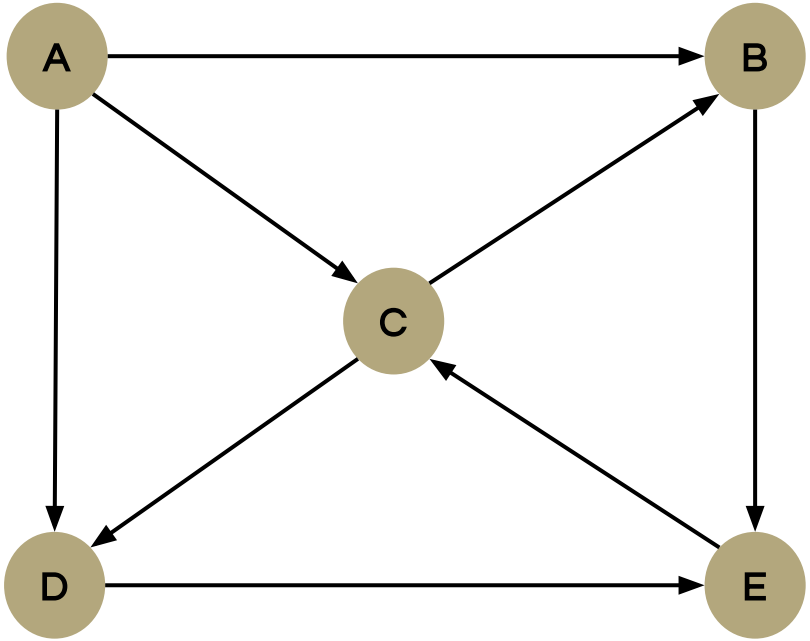
Breadth First Search

- **DFS Time Complexity:**
  - Using Adjacency List: O(V + E)
  - (V = vertices, E = edges)
  - Using Adjacency Matrix: $O(V^2)$
- **DFS Space Complexity: O(V)**
  - (for visited array + queue)

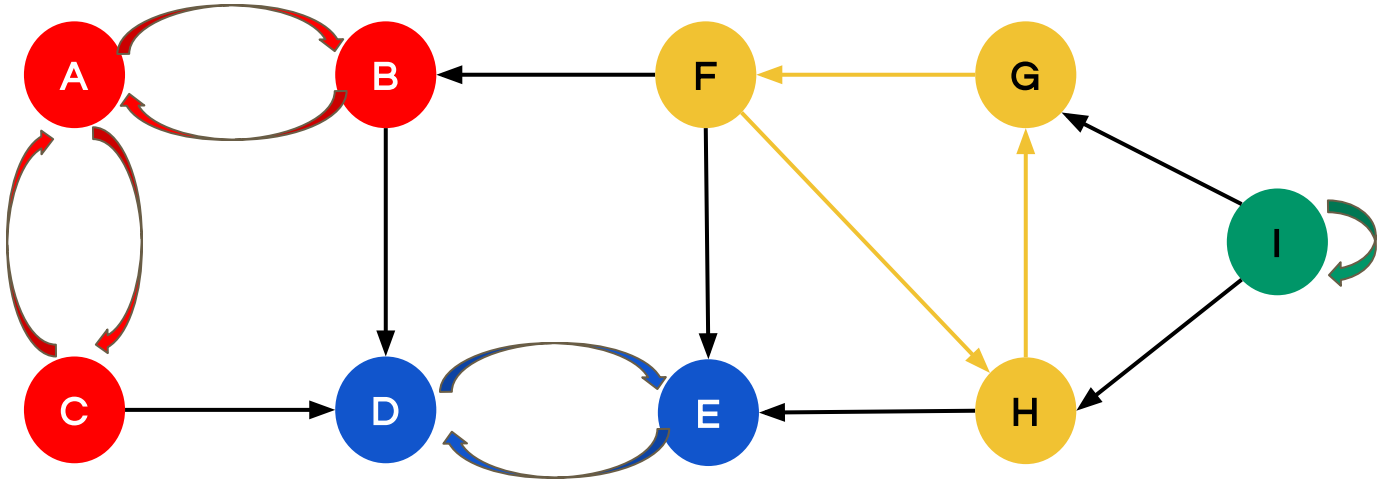


Applications of Breadth First Search

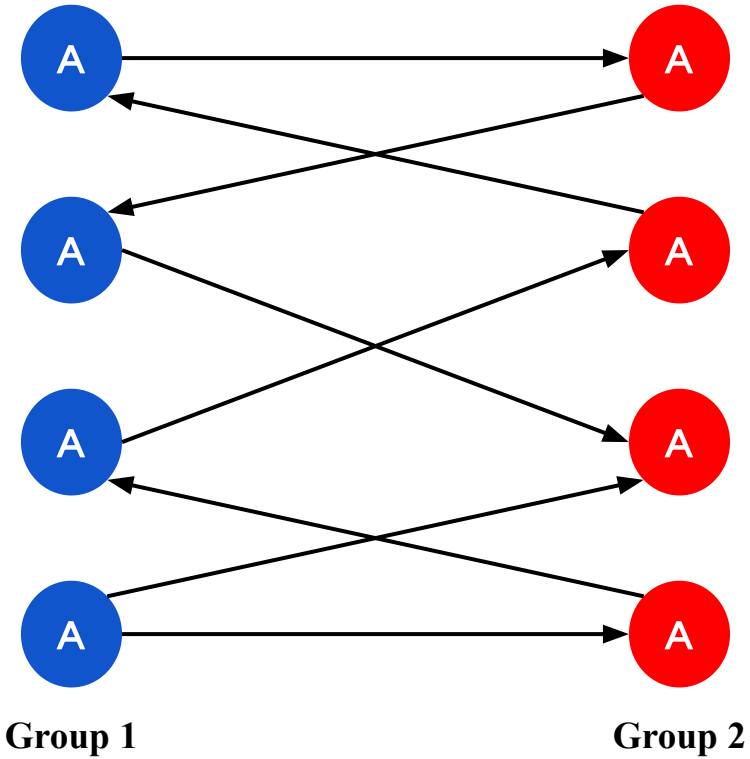
Detecting Cycle in a Directed Graph



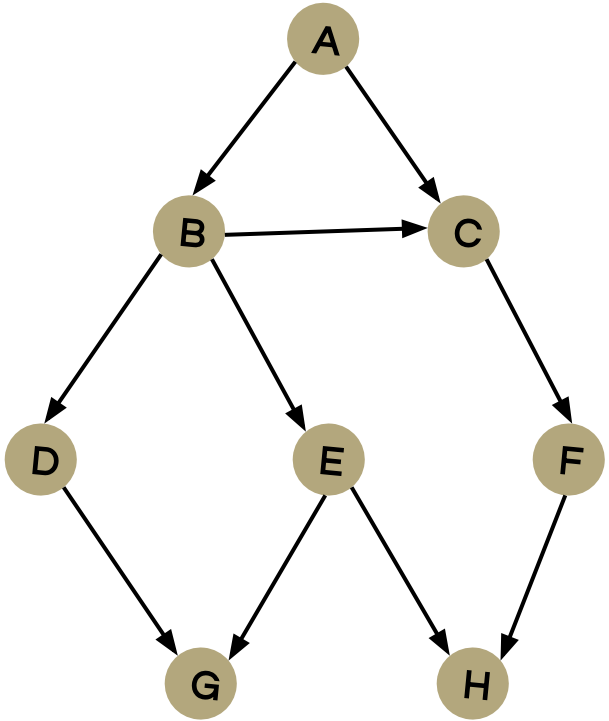
Finding Strongly Connected Components of a graph



To test if a graph is Bipartite

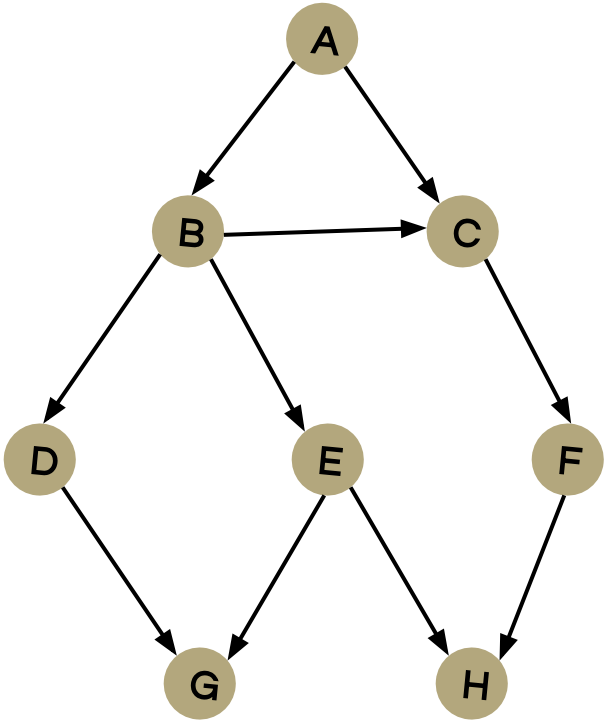


Breadth First Search



Vertex	Adjacent List
A	B, C
B	C, D, E
C	F
D	G
E	G, H
F	H
G	
H	

Breadth First Search(1)

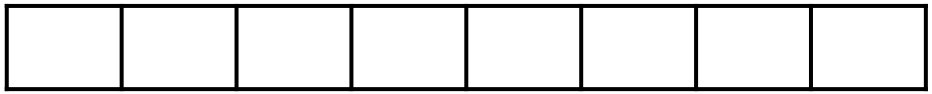


FRONT = 0



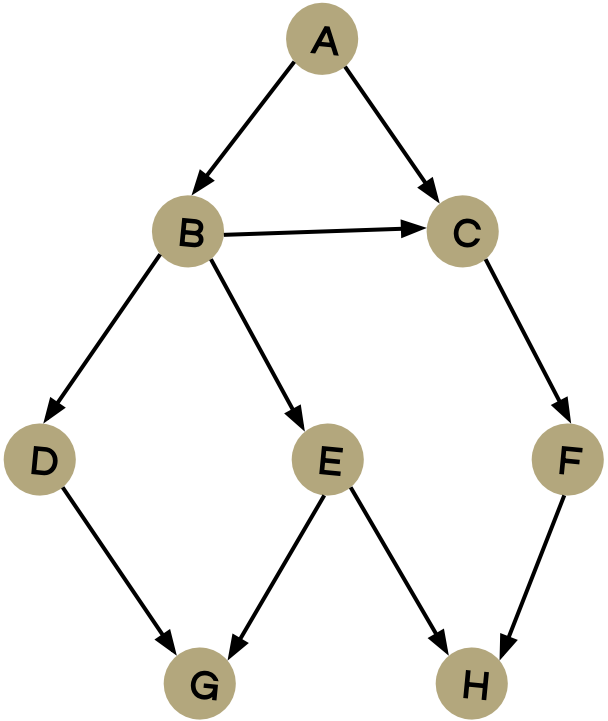
QUEUE

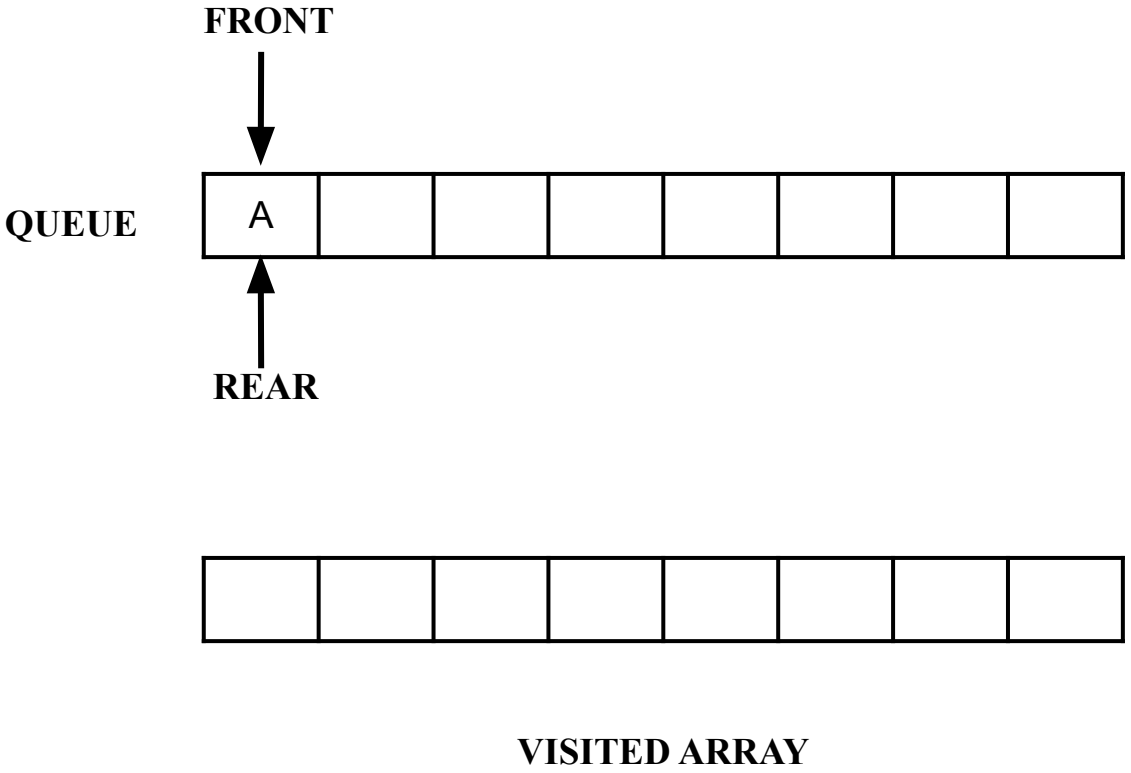
REAR = 0



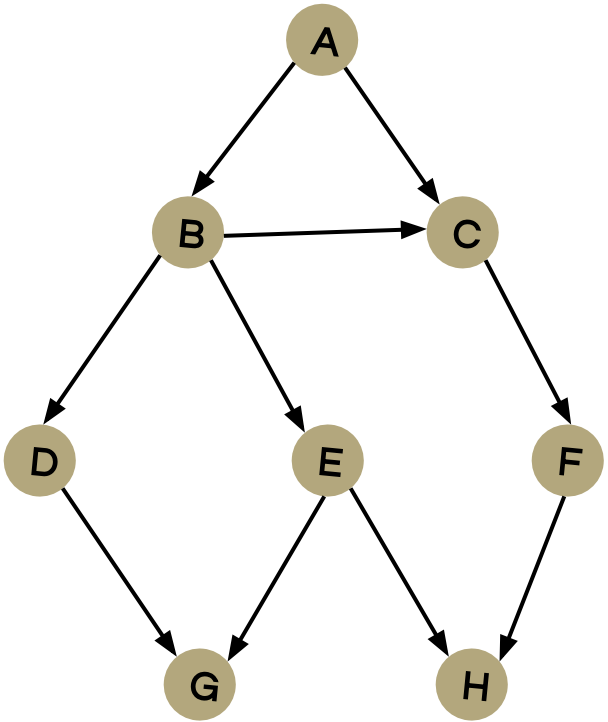
VISITED ARRAY

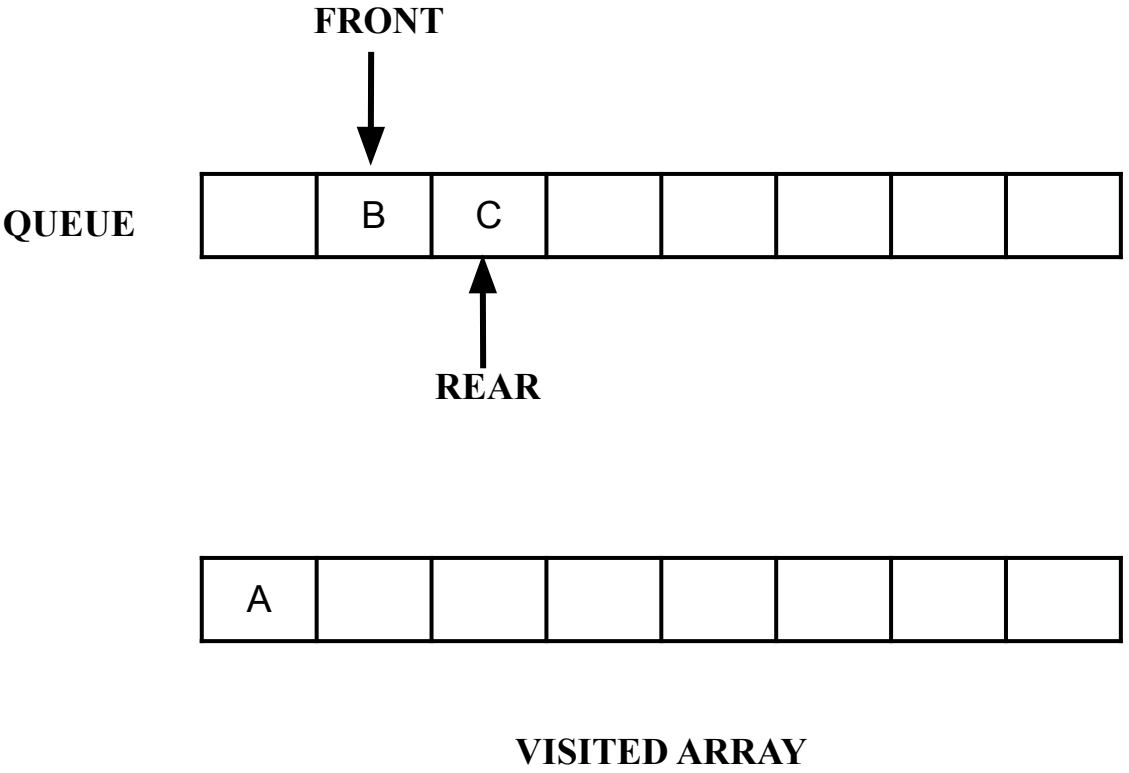
Breadth First Search(2)



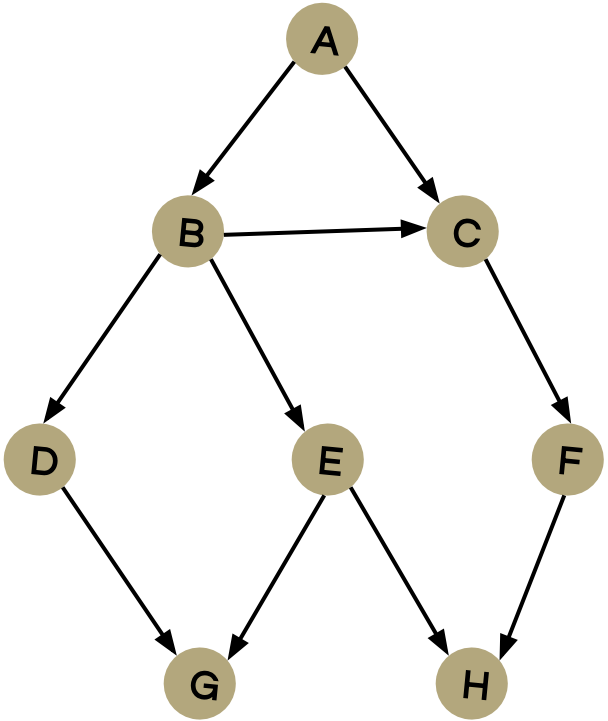


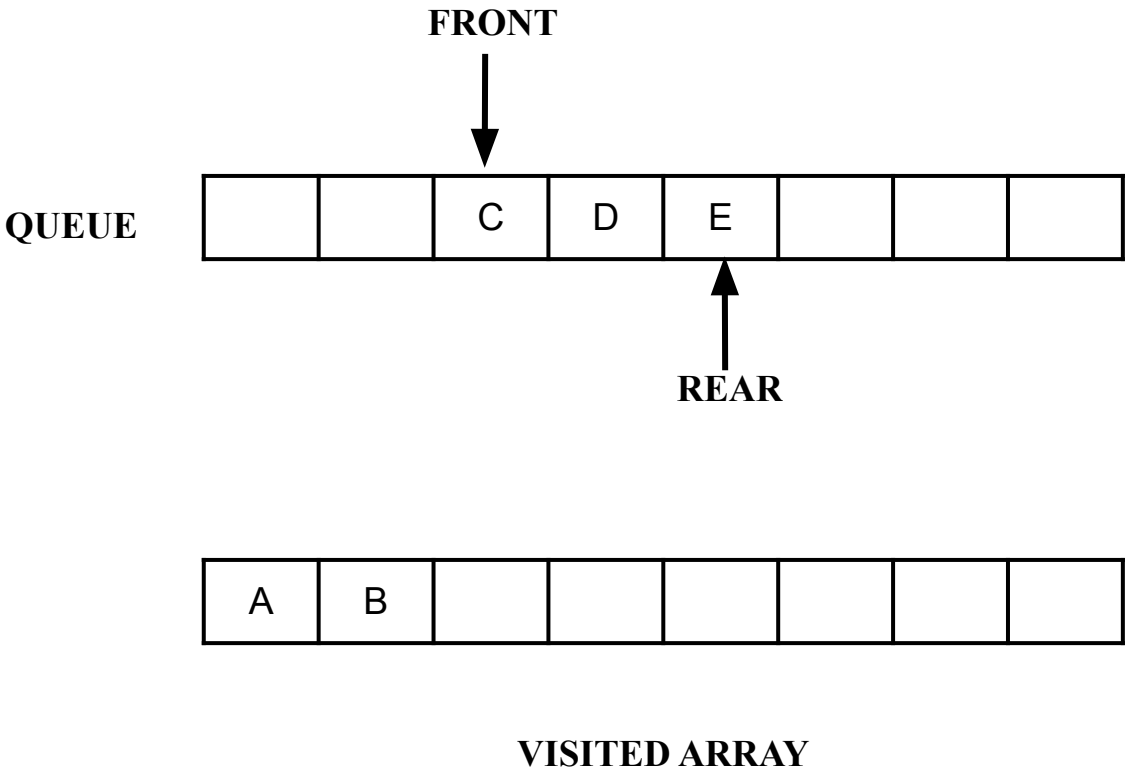
Breadth First Search(3)



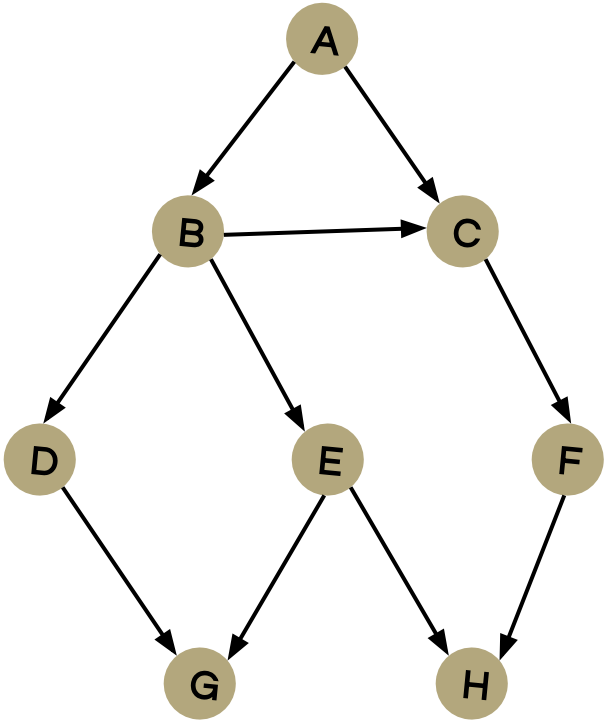


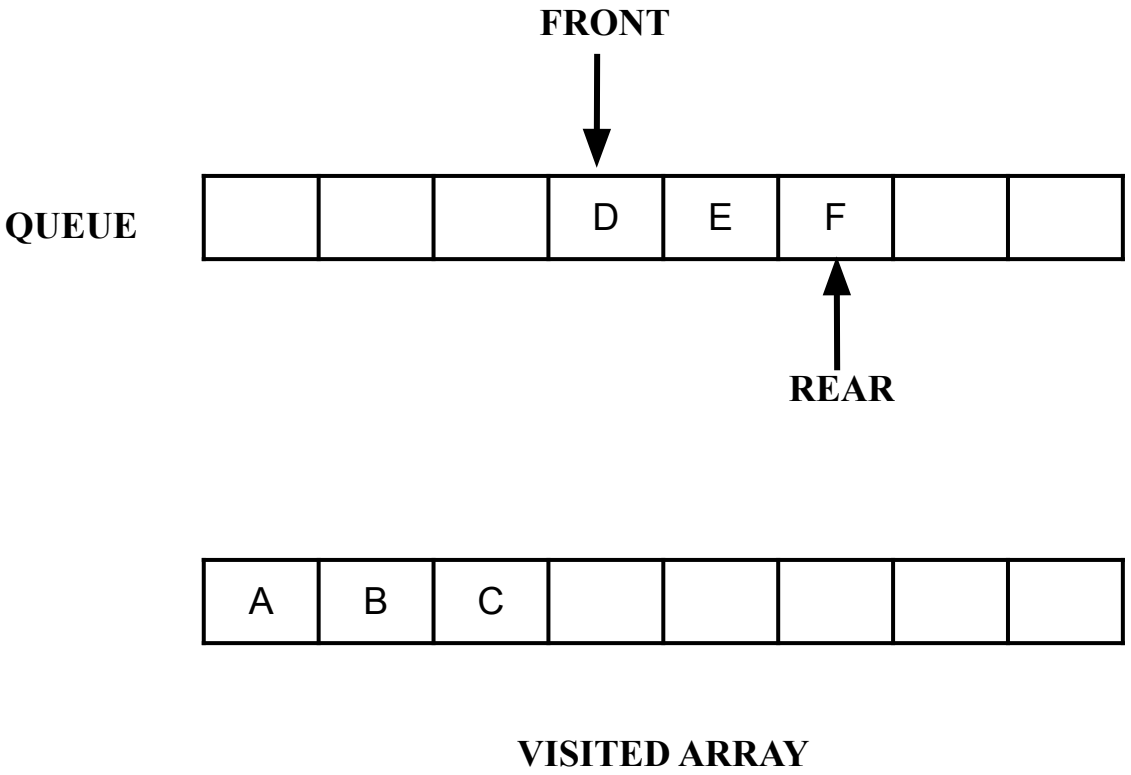
Breadth First Search(4)



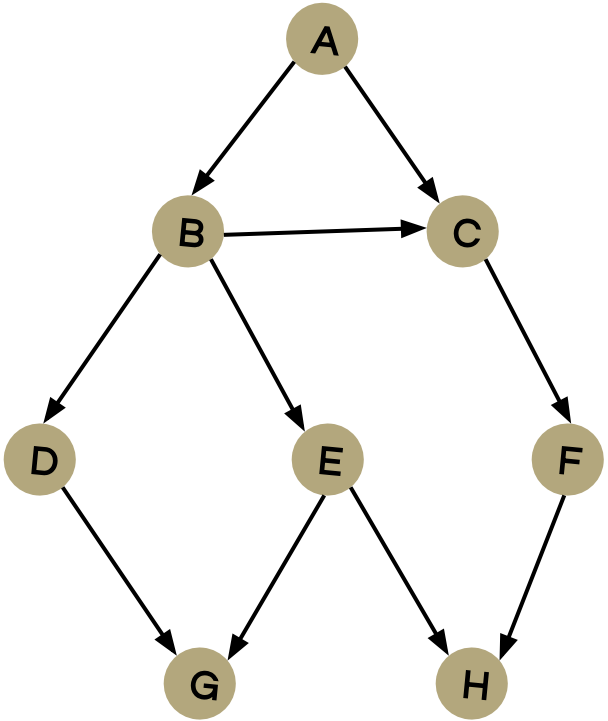


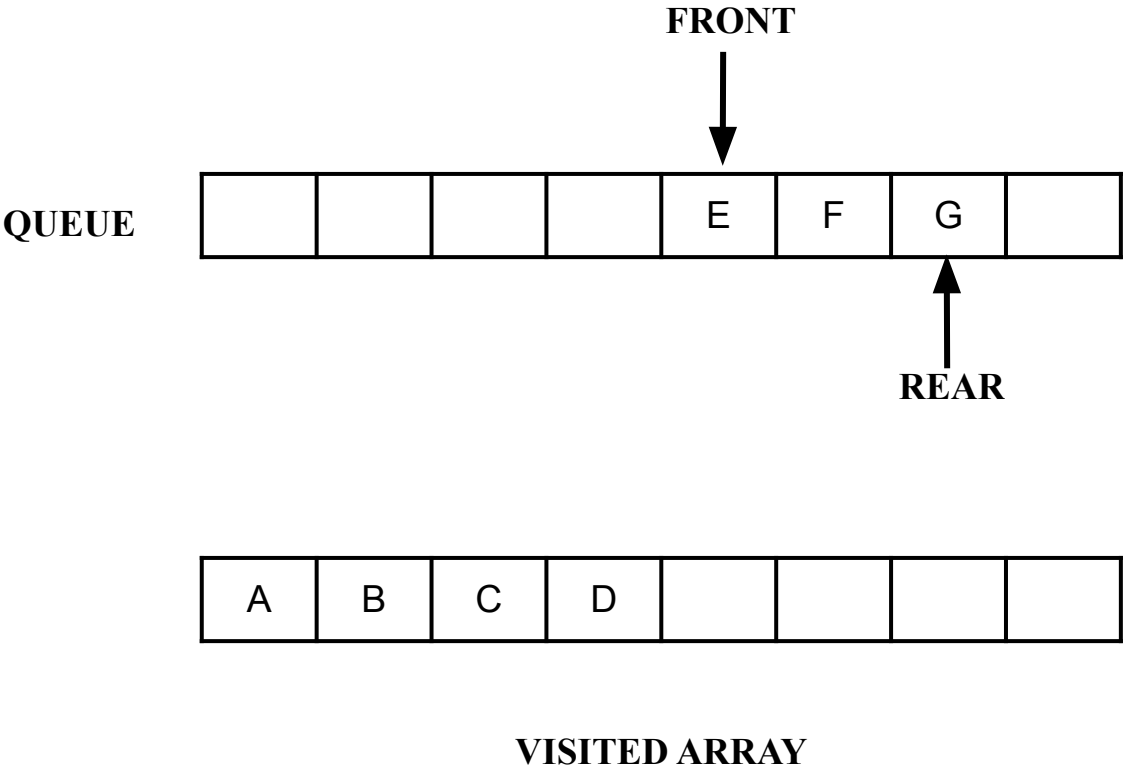
Breadth First Search(5)



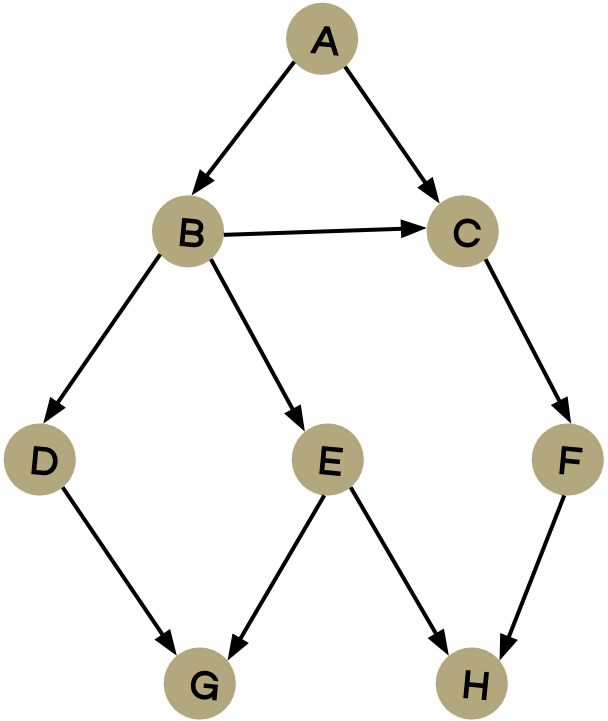


Breadth First Search(6)

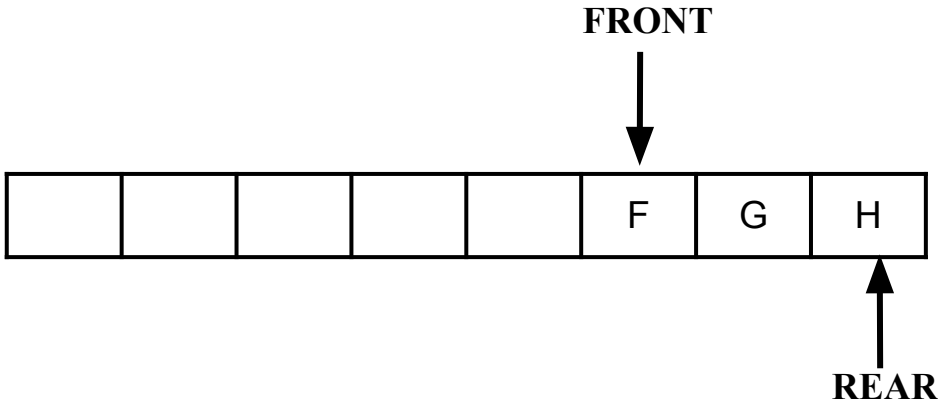




Breadth First Search(7)



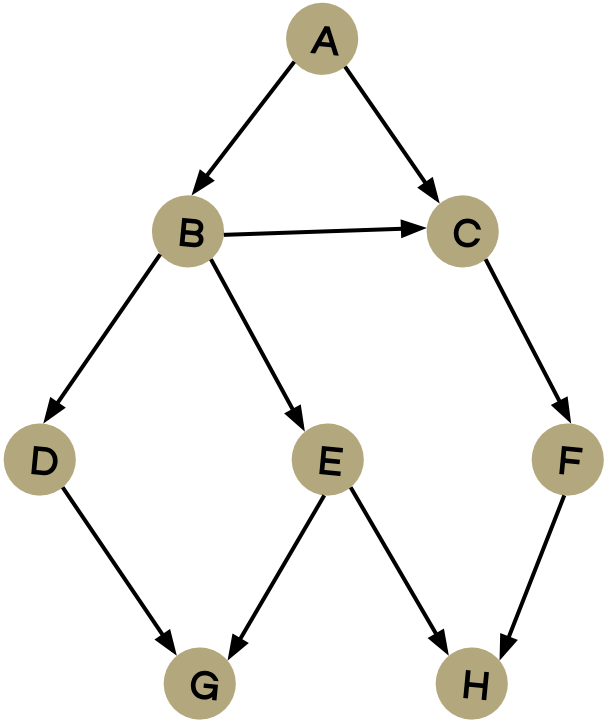
QUEUE



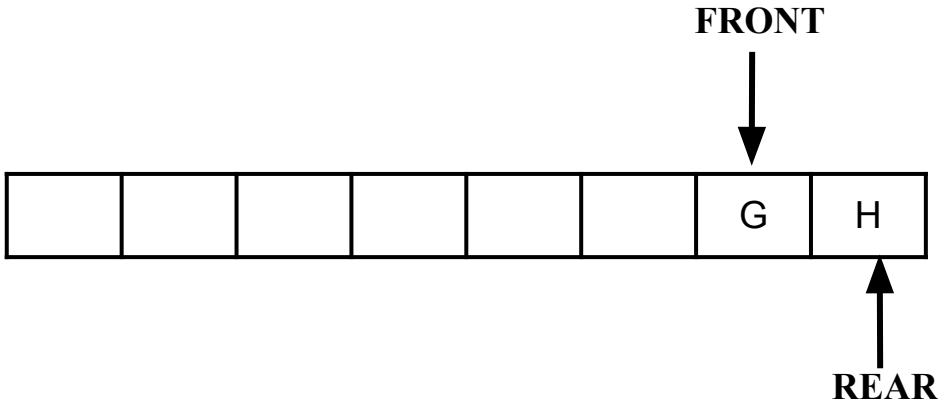


VISITED ARRAY

Breadth First Search(8)



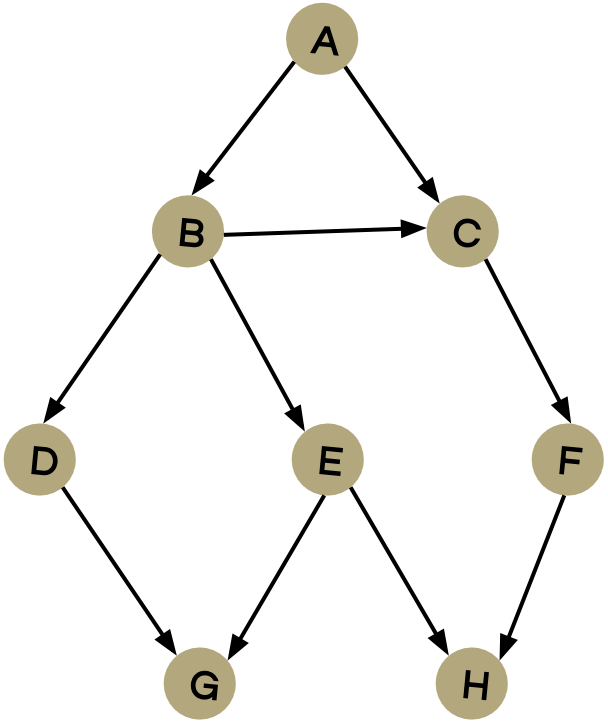
QUEUE



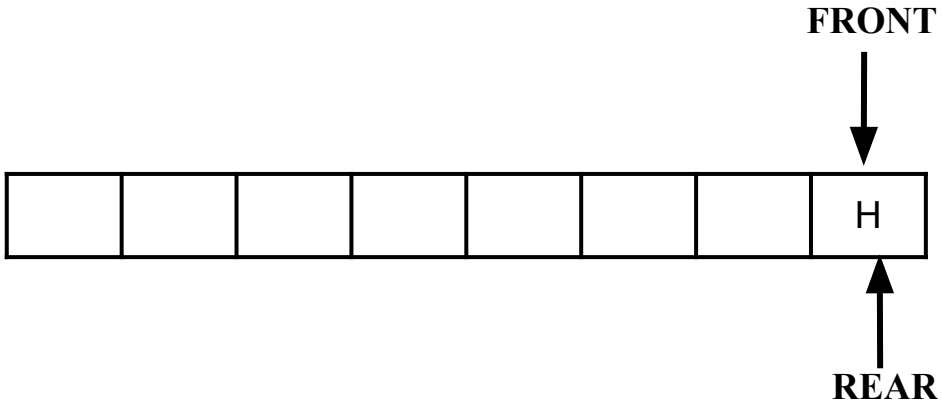


VISITED ARRAY

Breadth First Search(9)



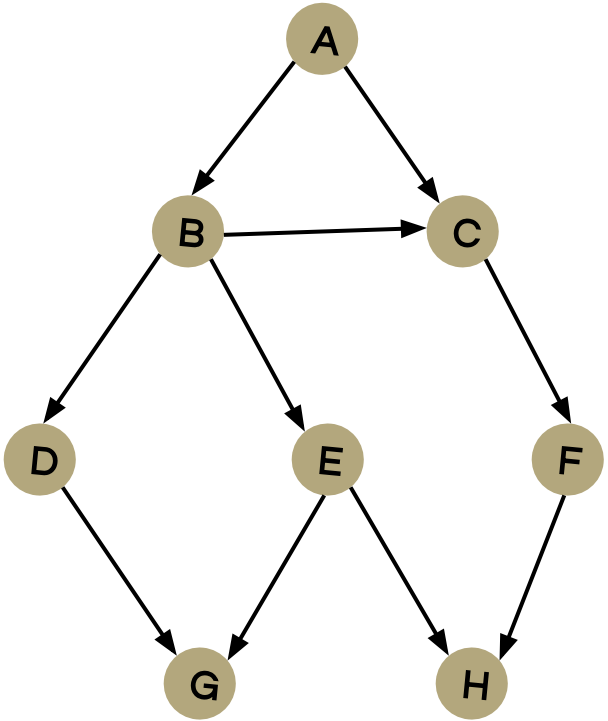
QUEUE





VISITED ARRAY

Breadth First Search(10)



FRONT

QUEUE

							H

REAR

A	B	C	D	E	F	G	H

VISITED ARRAY

Breadth First Search

Advantages :

- Guaranteed to find a solution if it exists.
- Finds the shortest/optimal solution in unweighted graphs.
- Will not get stuck in infinite paths.

Disadvantages :

- **High memory usage as entire levels are stored in memory.**